

Experiment: 6

5G NR Frame Structure Analysis with Different Subcarrier Spacing

Aim

To analyze the 5G NR frame structure by studying the effect of different subcarrier spacings, slot durations, and OFDM symbols using MATLAB.

Objectives

1. To analyze different Subcarrier Spacing (SCS) values used in 5G NR numerology.
2. To compare the slot duration corresponding to different Subcarrier Spacing values in 5G NR.
3. To study the number of OFDM symbols present in a 5G NR slot for normal cyclic prefix.

Software Required

- MATLAB

Theory

Unlike previous generations of mobile communication, 5G New Radio (NR) employs flexible numerology, where different subcarrier spacings are used depending on the deployment scenario

and service requirements. Increasing the subcarrier spacing reduces the slot duration, enabling lower latency and supporting diverse applications such as enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC), and massive Machine-Type Communications (mMTC). Regardless of the selected numerology, a normal cyclic prefix slot contains 14 OFDM symbols.

Objective – 1

Analyze the standard subcarrier spacing (SCS) values defined in the 5G NR numerology using MATLAB. Compare the relationship between the numerology index (μ) and subcarrier spacing to understand the flexible bandwidth allocation employed in 5G communication systems.

Procedure

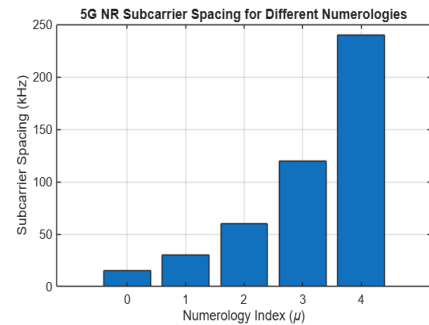
- Define the supported 5G NR subcarrier spacing values.
- Display the values using MATLAB.
- Plot the subcarrier spacing for different numerologies.

Output

```

1  clc;
2  clear;
3  close all; scs = [15 30 60 120 240]; mu = 0:4; bar(scs)
4  grid on
5  set(gca,'XTick',1:5)
6  set(gca,'XTickLabel',mu)
7  % Subcarrier Spacing (kHz)
8  % Numerology Index
9  xlabel('Numerology Index (\mu)')
10 ylabel('Subcarrier Spacing (kHz)')
11 title('5G NR Subcarrier Spacing for Different Numerologies')

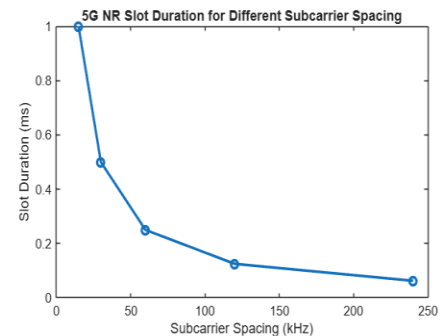
```



```

1  clear;
2  close all;
3  scs = [15 30 60 120 240]; slot = [1 0.5 0.25 0.125 0.0625]; % Subcarrier Spacing (kHz)
4  % Slot Duration (ms)
5  plot(scs, slot, 'o-', 'Linewidth', 2)
6
7  xlabel('Subcarrier Spacing (kHz)')
8  ylabel('Slot Duration (ms)')
9  title('5G NR Slot Duration for Different Subcarrier Spacing')

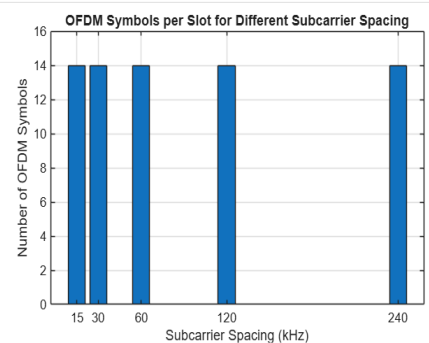
```



```

1  clc;
2  clear;
3  close all;
4  scs = [15 30 60 120 240]; symbols = [14 14 14 14 14]; figure
5  bar(scs, symbols)
6  % Subcarrier Spacing (kHz)
7  % OFDM Symbols per Slot
8  grid on
9  xlabel('Subcarrier Spacing (kHz)')
10 ylabel('Number of OFDM Symbols')
11 title('OFDM Symbols per Slot for Different Subcarrier Spacing')
12 ylim([0 16])

```



Result

The MATLAB analysis successfully demonstrated the flexible numerology of 5G NR. The results

showed that increasing the subcarrier spacing decreases the slot duration while maintaining a constant number of OFDM symbols per slot under the normal cyclic prefix configuration.